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Static, kinetic rest charge localisation (conservative system):

...

Stationary, slow charge movement localisation (conservative system):

$$\begin{aligned} \nabla \cdot \varepsilon_0 \mathcal{E} = \varrho & \quad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t} \mathcal{B} = i\omega \mathcal{B} & \quad \nabla \times \frac{1}{\mu_0} \mathcal{B} = j + \varepsilon_0 \frac{\partial}{\partial t} \mathcal{E} = j - i\omega \varepsilon_0 \mathcal{E} \\ & & (1) \end{aligned}$$

Dynamic, fast charge movement (relativistic):

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